# JOURNEY RISK MANAGEMENT (JRM) STUDY

## Salem Terminal to SANKAGIRI LORRY OWENER'S ASSOCIAATI

## Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

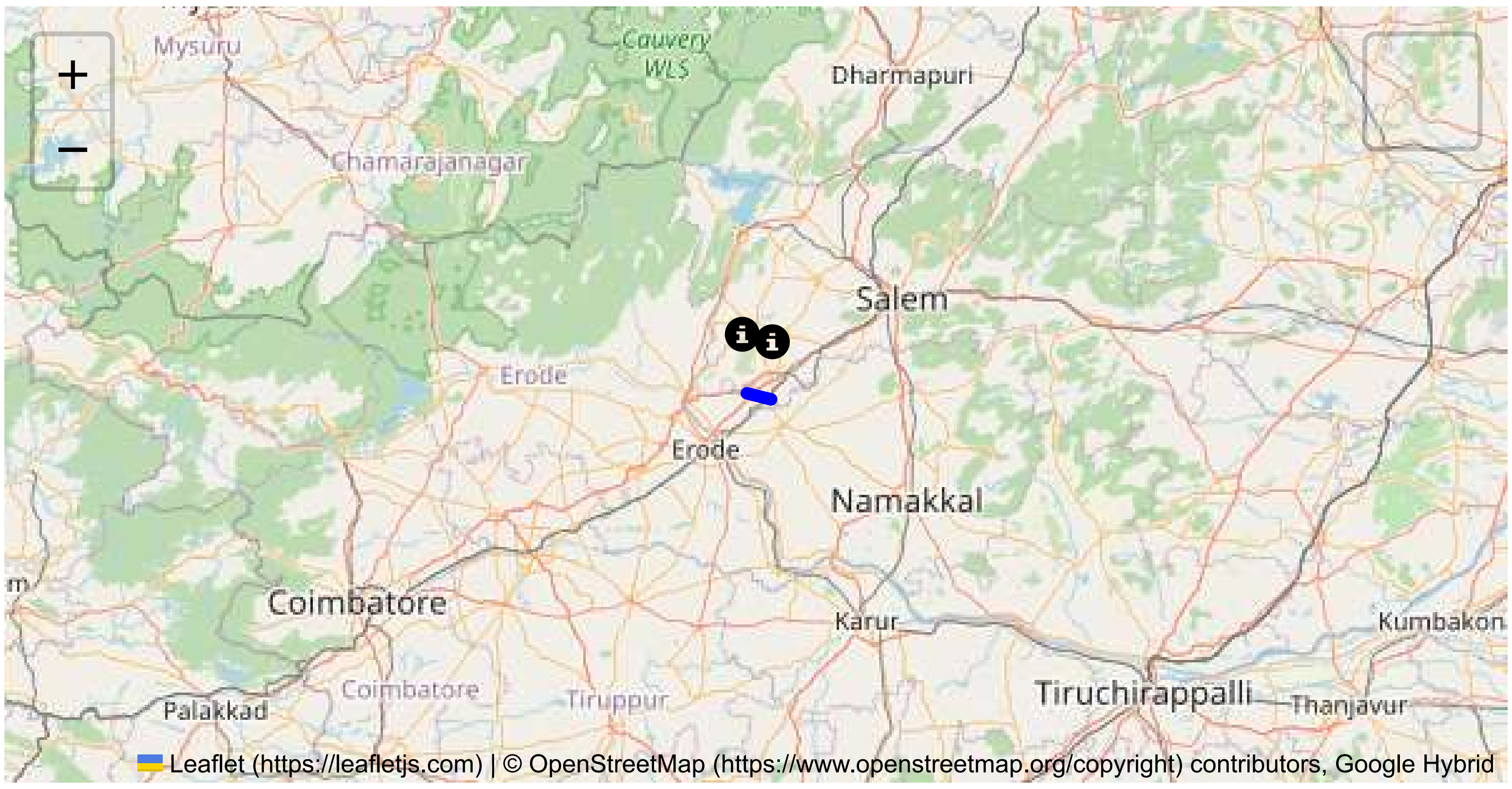
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

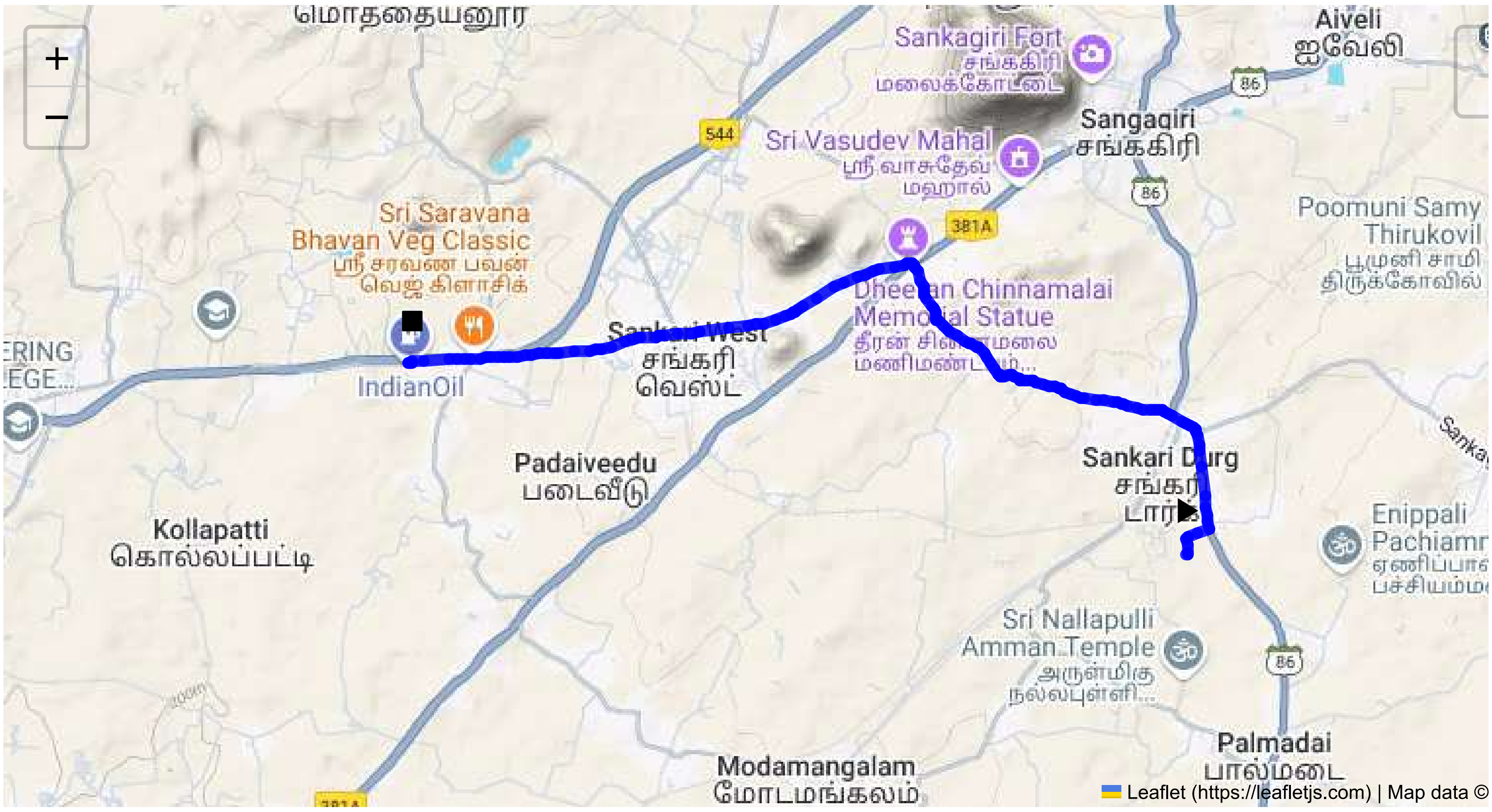
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.




## Route Summary:
## Total Distance: 9.82 km
## Estimated Duration: 0.3 hours
## Adjusted Duration (Heavy Vehicle): 0.3 hours
## Start: (11.4381, 77.8734)
## End: (11.45446346, 77.80517969)




# Welcome to the Journey Risk Management Study

Route Safety Analysis: CVQF+23W, Sangagiri to FR34+R2F, Virachchipalayam

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1. **Overview of the Route Map:**

- The route spans approximately 9.82 kilometers in Tamil Nadu, primarily taking local roads that connect small towns and rural areas. The journey typically follows arterial roads leading through agricultural and semi-urban regions.

2. **Typical Weather Conditions and Potential Weather-related Hazards:**

    - The region experiences hot, dry summers with temperatures often exceeding 35°C, and moderate to heavy rainfall during the monsoon season (June to September).
    - Potential weather-related hazards include slippery roads during monsoons, decreased visibility in heavy rain, and high temperatures that could affect vehicle operations.

3. **Analysis of Traffic Patterns:**

    - Traffic is likely to peak during early morning (8:00-10:00 AM) and late afternoon (4:00-6:00 PM) as residents commute.
    - Roads leading into and out of Sangagiri and Virachchipalayam can experience congestion, especially near marketplaces and intersections.

4. **Assessment of Road Quality and Infrastructure:**

    - Roads are generally paved but may have sections with potholes and uneven surfaces, especially post-monsoons.
    - Some stretches of the road may lack clear lane markings, and there may be inadequate street lighting in certain areas.

5. **Suggestions for Alternative Routes for Emergencies:**

    - Alternate routes may include taking secondary roads that bypass congested areas, though these may be narrower and less maintained.
    - Coordinates with local traffic management resources for real-time updates on detours.

6. **Summary of Local Regulations Affecting Hazardous Material Transport:**

    - Transport requires adherence to Tamil Nadu's regulations on hazardous materials, including specific permits and time restrictions (usually avoiding peak public movement periods).
    - Vehicles must display proper signage and carry necessary safety equipment.

7. **Overview of Historical Incidents:**

    - While specific historical data on incidents might not be widely documented, local reports suggest occasional accidents involving trucks, primarily due to speeding or inadequate road conditions.

8. **Environmental Considerations and Sensitive Areas:**

    - The route passes near agricultural lands, where spills could affect soil and water quality.
    - Ensure compliance with regulations protecting local wildlife and wetlands.

9. **Analysis of Communication Coverage:**

    - Most of the route has adequate mobile network coverage, but potential dead zones may exist in remote rural segments.
    - It is advisable to have offline maps and alternative communication tools.

10. **Estimated Emergency Response Times:**

- In urban areas like Sangagiri, response times are typically around 15-30 minutes depending on traffic.
- In rural stretches, the response time can extend to 45 minutes due to distance from emergency services.

11. **Overall Summary of Risk Assessment:**

    - The primary risks include weather conditions, road quality, and congestion during peak hours.
    - It is crucial to drive cautiously, particularly in inclement weather and adhere strictly to hazardous material transport regulations.
    - Preparedness for emergency detours and contingency plans in communication and vehicle breakdowns is essential.

By considering these factors, drivers can significantly mitigate risks along this route, ensuring safe and efficient transportation of hazardous materials.

## Risk Assessment - Turns

| | Risk Type | Risk Level | Coordinates | Speed Limit |
|---|---|---|---|---|
| 0 | Turn | High | 11.43811, 77.87348 | 15 KM/Hr |
| 1 | Turn | Medium | 11.43956, 77.87340 | 30 KM/Hr |
| 2 | Turn | Medium | 11.43971, 77.87348 | 30 KM/Hr |
| 3 | Turn | High | 11.44029, 77.87544 | 15 KM/Hr |
| 4 | Turn | Medium | 11.46307, 77.84941 | 30 KM/Hr |
| 5 | Turn | Medium | 11.46318, 77.84913 | 30 KM/Hr |
| 6 | Turn | High | 11.45517, 77.81345 | 15 KM/Hr |
| 7 | Turn | High | 11.45521, 77.81343 | 15 KM/Hr |
| 8 | Blind Spot | Blind Spot | 11.45481, 77.80540 | 10 KM/Hr |
| 9 | Turn | High | 11.45465, 77.80541 | 15 KM/Hr |
| 10 | Blind Spot | Blind Spot | 11.45464, 77.80498 | 10 KM/Hr |

## Emergency Locations

## Crowded Spots

| | type | name | coordinates | speed_limit | risk_level |
|---|---|---|---|---|---|
| 0 | school | KRP Matric. Hr. Sec School | 11.4546193, 77.8142445 | 30 km/h | Medium |

## Route Photos of Risky Spots


**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44029, 77.87544


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46307, 77.84941

**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46318, 77.84913

**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45517, 77.81345

**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45521, 77.81343


**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.45481, 77.80540


**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45465, 77.80541


**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.45464, 77.80498